

ferent signals in the same fibre.

To understand WDM transmission, let us consider a highway analogy where one fibre can be thought of as a multi-lane highway. Traditional TDM systems use a single lane of this highway and increase capacity by moving faster on this lane. Optical networking utilising DWDM is analogous to accessing the unused lanes on the highway to gain access to an incredible amount of untapped capacity in the fibre.

An additional benefit of optical networking is that the highway is blind to the type of traffic that travels on it. Consequently, the vehicles on the highway can carry asynchronous transfer mode (ATM) packets, SDH and IP.

Such a network will bring intelligence and scalability to the optical domain by combining the intelligence and functional capability of SONET/SDH, the tremendous bandwidth of DWDM and innovative networking software to spawn a variety of optical transport, switching and management-related products. This, coupled with the availability of 32-channel and 96-channel WDM, has led to the development of systems capable of 320Gbps to 1.2 terabit-per-second transmission.

Evolution of DWDM

Early WDM began in the late 1980s using the two widely spaced wavelengths in the 1310nm and 1550nm (or 850nm and 1310nm) regions, sometimes called wideband WDM.

The early 1990s saw a second generation of WDM, sometimes called narrowband WDM, in which two to eight channels were used. These channels were spaced at an interval of about 400GHz in the 1550nm window. Traditional passive WDM systems were widespread with 2-, 4-, 8-, 12- and 16-channel counts. This technique usually has a distance limitation of less than 100km, without any repeater stage. Another variant of WDM, known as coarse WDM (CWDM), typically used 20nm spacing of up to 18 channels that could reach a target distance up to about 50km on single-mode fibres.

By the mid-1990s, dense WDM

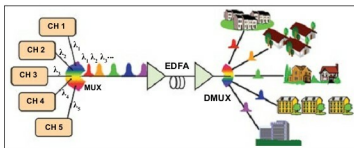


Fig. 2. Block diagram of a DWDM system

(DWDM) systems were emerging with 16 to 40 channels and spacing from 100GHz to 200GHz. By the late 1990s, DWDM systems were capable of 64 to 160 parallel channels, densely packed at 50GHz or even 25GHz intervals at distances of several thousand kilometres with amplification and regeneration along the route.

Interestingly, in a WDM network, the vast optical bandwidth of a fibre (approximately 30THz) is carved up into wavelength channels, each of which carries a data stream individually. The multiple channels of information (each having a different carrier wavelength) are transmitted simultaneously over a single fibre (Fig. 2).

The reason why this can be done is that optical beams with different wavelengths propagate without interfering with one another. When the number of wavelength channels is above 20 in a WDM system, it is generally referred to as dense WDM or DWDM.

Optical fibre transmission uses wavelengths that are in the near-infrared portion of the spectrum, just above the visible and thus undetectable to the unaided eye. There are ranges of wavelengths known as transmission windows at which the fibre operates best (Table II).

These wavelengths best match the transmission properties of available light sources with the transmission qualities of optical fibre. Usually, LEDs are used for 850nm or 1310nm multi-mode applications and lasers are used for 1310nm or 1550nm single-mode applications.

For DWDM system, 1550nm window has been chosen due to low-attenuation characteristics of glass at 1550nm and the fact that erbium-

doped fibre amplifiers (EDFAs) operate in the 1530nm-1570nm range. The International Telecommunications Union (ITU) defines a frequency grid that specifies standard frequency spacing as

100GHz, which translates into a 0.8nm wavelength spacing.

DWDM system components

Transmitter and receiver. A transmit transponder is used at transmitter's side to change electrical signals into ITU grid-specific optical wavelengths. For this purpose, it employs a narrow band of laser to generate optical pulses. At the receiver's side, a receive transponder converts optical pulses to electrical signal, if required. In these systems, small form-factor pluggable (SFP) transceivers are used to interface the system with the fibre-optic cable.

Transmission media. Due to minimum fibre-optical attenuation at 1550nm and availability of optical amplifiers in 1550nm region, most of the DWDM systems operate in C-band, that is, from 1530nm to 1565nm. In this range, conventional single-mode optical fibres (ITU G.652 compliant) undergo high dispersion, which severely limits the distance between repeater stations. Solutions lie in the use of dispersion compensating fibre (DCF) and zero dispersion fibre (ITU G.653 compliant).

But, DCF adds additional power loss and zero dispersion fibre results in nonlinearity such as four-wave mixing (FWM). A fibre that has small but non-zero amount of dispersion can minimise the non-linearity effects. The ITU G.655 compliant, non-zero dispersion fibre (NZDF) has dispersion which is carefully chosen to be small enough to enable high-speed transmission over long distances, but large enough to suppress FWM. So, ITU G.652 is preferred for access network and ITU G.655 for backbone network. In summary, a future-proof fibre-optic DWDM network should have a combination of ITU G.652